Lesson Plan: Kohiki Slip Decoration for Slab Building

Objectives

- Introduce participants to the history and aesthetics of kohiki slip decoration.
- Teach slab-building fundamentals with emphasis on surface treatment.
- Guide students through applying kohiki slip and integrating it into functional or sculptural forms.

Materials

- Clay body (dark stoneware recommended).
- White slip (prepared to a thick cream consistency).
- Brushes, sponges, etc…
- Straight edges, rulers, tar paper for patterns.
- Rolling pins or slab roller.
- Canvas boards for rolling slabs.
- Wooden boards for working surfaces.
- Bevel cutters, knives, and scoring tools.
- Reference images of kohiki ware.

Activities

1. **Introduction (20 min)**: Brief history of kohiki slip; show examples.
2. **Slab Preparation (40 min)**: Demonstrate rolling slabs evenly; discuss thickness control.
3. **Slip Application (40 min)**: Demonstrate brushing, dipping, or pouring slip; encourage freedom and experimentation.
4. **Lunch and Patience (≈1.5 hr)**: Allow time for slip and clay to firm up.
   - **Structured Lunch/Practice Slot (20–30 min)**: Provide short guided exercises students can do while waiting.
   - **Goal**: Turn downtime into deliberate practice focused on slip observation and simple brushwork.
5. **Discussion on Leatherhard Stage (20 min)**: Talk about ideal leatherhard conditions for construction; show examples.
6. **Slab Construction (1 hr)**: Demonstrate joining slabs into simple forms; emphasize scoring and compression.
7. **Wrap-Up Discussion (15 min)**: Share results; discuss how kohiki surfaces change in firing.

Teaching Notes

- Stress patience: kohiki slip rewards subtlety.
- Encourage integration of surface decoration with form.
- Highlight how firing atmosphere alters slip appearance.

Kohiki Slip

Timeline: History of Kohiki Slip

1392–1910 Korean Yi Dynasty Punch'ŏng ware: iron-rich clay with white slip → Late 1500s Introduction to Japan Korean potters bring slip techniques → 16th–17th c. Japanese Tea Ceremony Kohiki valued for wabi-sabi aesthetics → 20th–21st c. Modern Revival Contemporary artists adapt kohiki in wood/soda firing

Parallel Timeline: Import of Korean and Chinese Potters to Japan

1592–1598 Imjin Wars Toyotomi Hideyoshi invades Korea → Late 1500s–early 1600s Forced Migration Thousands of Korean potters relocated to Japan → 17th c. Transmission of Techniques Punch'ŏng slip, porcelain, celadon introduced → 17th–18th c. Integration into Japanese Ceramics Kohiki, Arita porcelain, and hybrid styles emerge

Late 1500s–early 1600s Forced Migration Thousands of Korean potters relocated to Japan ⇢ Late 1500s–early 1600s Shino Ware Development Mino region, Japan's first white glaze pottery, parallel to Kohiki ⇢ 17th–18th c. Integration into Japanese Ceramics Kohiki, Arita porcelain, and hybrid styles emerge

A couple modern examples of popular kohiki slip decoration by Akira Satake and Robert Fornell, as shown in the images below. I started experimenting with kohiki slip decoration after seeing Satake's work.

A beautiful example of a more traditional kohiki piece is this tea bowl identified as Agano ware at this website: `https://qualitychanoyu.com/2023/01/13/antique-agano-ware-kobiki-chawan/`

Antique Agano Shiro-Kohiki Chawan

Other links of interest about kohiki ware:

- `https://turuta.jp/ceramics-story/archives/21164`
- `https://dawan-chawan-chassabal.blogspot.com/2010/02/powdery-matsudaira-at-hatakeyama-museum.html`
- `https://dawan-chawan-chassabal.blogspot.com/2010/11/powdery-matsudaira-revisited.html`

Lunch / Patience: Guided Exercises

Add discussion here about below material!

Guided Practice (20–30 min)

Short, focused exercises. Keep materials minimal: work on wooden bats for slip applications.

1. **Single-Stroke Studies (6 min)**: Make one confident stroke across the bat–experiment with pressure and angle.
2. **Opacity Ladder (6 min)**: Apply 3 horizontal bands with decreasing slip thickness (heavy → medium → thin). Let dry and note differences.
3. **Quick Pair Share (6 min)**: Swap bats with a partner; each gives one suggestion for the other's next step.
4. **Identify Area for Cutting and Construction (6 min)**: On your bat, mark an area you might cut out later for slab construction; consider how the slip application will interact with form.

Add discussion here about below material!

Slip Consistency & Quick Recipe

- Basic Kohiki Slip Recipe:

EPK kaolin:	500 g
OM4 ball clay:	400 g
G-200 feldspar:	600 g
Silica:	500 g

- Water: variable–depends on deflocculant use if any.
  1. Thin toothpaste consistency
  2. Thick cream consistency
  3. Any range in between depending on your application style and decorative goals
- Deflocculant (sodium silicate or Darvan): 1–3 g, if needed to thin without settling[1]

Add discussion here about below material!

Quick Tests

- **Brush-Load Test**: Dip brush and make a 10 cm stroke; note opacity and pooling.
- **Adhesion Test**: Apply slip to a scrap of the class body, bend gently when leatherhard—does it crack or flake?
- **Drying Rate**: Make a thin swatch and time until it is no longer tacky—record for next session.
- **Hypothesis**: Suggest one hypothesis on slip application and test.

Brushwork Patterns (Step-by-step)

Keep demonstrations short; show one pattern then let students repeat.

1. **Stippling**: Load brush, dab vertically using the tip only; vary density for tonal control.
2. **Drag-and-Stop**: Pull brush in a steady stroke, lift quickly for crisp ends.
3. **Combing**: Use a fork or comb dragged through semi-wet slip for parallel texture.
4. **Ribs**: Drag a rib or flat tool through wet slip to create texture; vary angle and pressure for different effects.

Observation Prompts (for students)

Ask students to note these during drying and return to them after leatherhard stage:

- Changes in color/value as the slip dries.
- Areas of pooling vs even coverage.
- Edges where slip has feathered or pulled away.
- Any textural interaction where slip meets tool marks.

Add discussion here about below material!

Two Micro-lectures (5–7 min each)

extbfMicro-lecture A — Aesthetics of Kohiki (5 min): Discuss wabi-sabi, the value of subtlety, and how controlled imperfection enhances form. Show 2–3 images and ask students to name one element they find compelling.

[1] Deflocculants help reduce the amount of water needed, but can reduce the workability of the clay beneath the slip. Use sparingly and test adhesion.

extbfMicro-lecture B — Kiln Transformations (5 min): Explain how wood/soda firing can change slip surface—ash deposition, reduction/oxidation effects, and how slip texture traps or resists glaze.

Safety & Handling Notes

- Always wash hands after handling slips containing deflocculants.
- Keep sanding and dry-scraping to a minimum; use wet cleaning to avoid airborne silica.
- Label slip containers; avoid cross-contamination between white and colored slips.
- For soda firing: follow kiln-specific SDS guidance for soda ash and wear PPE when handling solutions.

What I Notice / What I Would Change (Worksheet)

1. Sample ID: ______________________________
2. Observations (3 lines): ______________________________
3. One change I would make next time: ______________________________
4. Confidence (1–5): ______________________________

Lesson Plan: Wood-Fired Soda Firing

Objectives

- Review fundamental principles of firing across all types of kilns.
- Discuss the unique characteristics of wood firing.
- Explain the chemistry and effects of soda firing.
- Introduce participants to the principles of soda firing in a wood kiln.
- Teach preparation of work for soda atmosphere (wadding, glazing, placement).
- Explore how soda interacts with clay and slip surfaces.

Materials

- Bisque-fired student work (preferably with surfaces prepped for wood+soda).
- Wadding materials (alumina hydrate + kaolin).
- Kiln shelves, posts, protective gear.
- Wood for firing (species discussion: pine, oak, etc.).
- Soda mix (soda ash + baking soda + whiting, in water) – following the suggestions of Gail Nichols.

Activities

1. **Introduction (20 min)**: Explain soda firing chemistry; show examples.
2. **Preparation of Work (30 min)**: Demonstrate wadding placement; discuss glaze choices.
3. **Kiln Loading (45 min)**: Explain loading strategies; emphasize safety and teamwork.
4. **Firing Process (30 min overview)**: Walk through firing stages; demonstrate soda introduction; discuss temperature targets.
5. **Cooling & Unloading (Next Session)**: Explain cooling protocols; plan for unloading and critique.
6. **Wrap-Up Discussion (15 min)**: Anticipate results; encourage embracing variation and unpredictability.

Teaching Notes

- Stress safety: protective gear, teamwork, kiln etiquette.
- Highlight unpredictability as part of the aesthetic.
- Connect back to constructed works: how soda atmosphere transforms surfaces.

Suggested Flow

- Works construction.
- Bisque firing.
- Soda firing preparation: kiln prep, loading, firing overview.
- Kiln unloading, critique, reflection.